1 Planning (15 marks)

Defining the problem (2 marks)

P *k* is the independent variable and *h* is the dependent variable, or vary *k*, measure *h*. [1]

P Keep mass of object constant. [1]

Methods of data collection (4 marks)

M Labelled diagram (minimum two labels) showing object (mass) attached to cord and other end of cord fixed (e.g. stand and clamp or hook) and rule(r) drawn vertically next to cord. [1]

M Method of measuring mass e.g. balance / scales. [1]

M *k* = (weight or force) / extension or *mg* / *extension*; allow graphical methods.
Allow any subject e.g. $mg = k \times extension$. [1]

M Use of rule to measure *h* or maximum distance / length (fallen by the object).
Allow clear indication on diagram (i.e. dotted lines) linking distance *h* to rule.
Do not credit length of cord. [1]

Method of analysis (3 marks)

✓ Plot a graph of $\frac{(h-L)^2}{h}$ against $1/k$ [Allow $2/k$ or $2m/k$ or m/k] [1]

✓ *g* = gradient / 2m [gradient / *m* or gradient or gradient / 2] [1]

✓ Relationship is valid if the graph is a straight line passing through the origin. [1]

Additional detail (6 marks)

D Relevant points [6]
1 Keep starting point constant/drop object from same position / use of electromagnet to drop object / ensure mass is dropped from fixed point / check object falls vertically
2 Rule(r) fixed e.g. retort stand
3 Method to determine extension, e.g. measure length of stretched cord and subtract original length / 50.0 cm. [Accept from a diagram]
4 Safety precaution linked to prevention of mass / cord hitting a person – use safety screen / goggles; sand tray to catch falling object if cord breaks
5 Trial experiment to locate approximate point of *h* / to prevent object hitting surface
6 Detailed use of video camera with slow motion or frame by frame playback / motion sensor clearly explained
7 Cord obeys Hooke's law or must not exceed elastic limit
8 Use set square to ensure ruler is vertical
9 For each cord, repeat experiment determine average *h*

Do not allow vague computer methods.

[Total: 15 marks]

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